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Astrocytic APOE3-Christchurch expression ameliorates brain amyloid-β pathology in 5xFAD mice

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The rare *APOE3*-Christchurch (APOE3Ch) variant is linked to resistance against *PSEN1* p.E280A-driven autosomal dominant Alzheimer's disease (AD). Recent studies in AD mouse models have demonstrated an effect of APOE3Ch in reducing tau pathology and tau propagation, yet its effects on amyloid pathology and related toxicity are not fully understood. While prior studies have reported reduced amyloid pathology with APOE3Ch, we extended this knowledge by investigating how astrocyte-specific expression of APOE3Ch impacts amyloid pathology and related responses in 5xFAD mice, an amyloid mouse model. Using adeno-associated virus (AAV)-mediated gene delivery, we overexpressed APOE3 or APOE3Ch in astrocytes of 5xFAD mice at the neonatal stage, then analyzed their effects during the advanced stage of amyloid pathology. Astrocytic APOE expression significantly reduced amyloid burden, neuritic dystrophy, and gliosis compared to GFP controls. Notably, astrocytic APOE3Ch expression, relative to APOE3, markedly lowered oligomeric Aβ levels and promoted the formation of more compact, fibrillar plaques, suggesting a shift toward a less toxic aggregation profile. Transcriptomic profiling of cortical tissue revealed broad downregulation of immune-related and proteostatic pathways. These findings indicate that astrocytic APOE3Ch sufficiently attenuates Aβ pathology and related toxicity, supporting its potential as a therapeutic modifier for AD.

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INTRODUCTION

Alzheimer's disease (AD) is a leading cause of dementia worldwide, characterized by amyloid-β (Aβ) deposition, neurofibrillary tangle formation, neuroinflammation, and progressive cognitive decline [1]. The *APOE* gene coding apolipoprotein E (APOE) is the greatest genetic modifier of AD risk with its isoforms differentially influencing Aβ pathology and associated neurotoxicity. While *APOE* ε4 allele (*APOE4)* is associated with the increased AD risk among the three gene alleles (ε2, ε3, ε4), *APOE* ε2 allele (*APOE2*) is protective compared to the reference *APOE* ε3 allele (*APOE3*) [2]. Of note, the rare *APOE* R136S *Christchurch* variant occurs in the *APOE* ε3 backbone (*APOE3Ch)* and has been reported in a *PSEN1* p.E280A carrier. Homozygosity for *APOE3Ch* in this individual resulted in markedly reduced tau pathology and delayed dementia onset despite extensive amyloid deposition [3]. This observation has sparked interest in investigating the potential protective mechanisms of *APOE3Ch* variant to advance AD therapeutic strategies. Recent studies in mouse models have confirmed the protective mechanisms of *APOE3Ch* variant. Knock-in expression of APOE3Ch in APP/PS1 amyloid model mice resulted in a modest reduction of amyloid pathology, a significant decrease in Aβ-driven tau seeding and spreading, and dampened microglial activation [4]. Moreover, introducing the Christchurch mutation into *APOE2*, *APOE4* or murine *Apoe* backgrounds resulted in decreased tau pathology and associated neuroinflammation in mouse models of tauopathy [5–7]. These findings collectively suggest that APOECh substitution potentially modulates amyloid pathology, tau pathology and propagation, and microglia-related inflammatory pathways. However, the extent to which these effects are cell-type specific remains poorly understood.

APOE is produced and secreted primarily by astrocytes in the brain, making astrocytes a prime target for studying the cell-type specific effects of APOE3Ch [8]. In this study, we investigated the impact of astrocyte-specific overexpression of APOE3Ch in the 5xFAD mouse model of amyloidosis. While APOE2Ch, APOE4Ch and murine APOECh models have also been reported, our study centered on APOE3*Ch*, the context of the original protective carrier, to isolate astrocytic effects in amyloid pathology. Genes coding APOE3, APOE3Ch, or a control GFP were delivered into the neonatal brains of 5xFAD mice using an AAV8-GFAP-driven expression system. Our results showed that astrocytic expression of *APOE3* and *APOE3Ch* leads to similar effects on Aβ pathology and gliosis. Nonetheless, APOE3Ch promoted the formation of more compact plaques, and reduced neuritic damage. These findings suggest that astrocytic APOE3Ch expression influences the aggregation state of Aβ differently compared to APOE3, but

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may be insufficient to broadly mitigate amyloid pathology and related toxicity compared to APOE3, highlighting the potential importance of its expression in other cell types, such as microglia. By focusing on the cell-type specificity of APOE3Ch's effects, this study raises important questions about its mechanisms of action and therapeutic potential in AD.

METHODS

Animals

5xFAD mice were acquired from the Jackson Laboratory and maintained as heterozygotes on a C57BL/6 genetic background. All animal experiments were conducted in compliance with the protocols of the Mayo Clinic Institutional Animal Care and Use Committee and the National Institutes of Health Guide for the Care and Use of Laboratory Animals. Both male and female mice were used in this study.

Viral vector construction and production

The AAV8 vector construction and production were carried out by the Hope Center Viral Vectors Core at Washington University School of Medicine. APOE3 or APOECh cDNA in the pcDNA3.1 vector was amplified using PCR, with forward primers incorporating a BamHI restriction site and reverse primers containing a SalI site along with an influenza hemagglutinin (HA) tag sequence. The PCR products were then digested with BamHI and SalI and inserted into an AAV8-GFAP-GFP backbone, where GFP was replaced with the respective PCR products. This process generated the constructs of AAV8-GFAP-APOE3-HA and AAV8-GFAP-APOE3Ch-HA, each encoding the corresponding APOE isoform with an HA tag under the control of the GFAP promoter.

The AAV8 viral particles were produced as described previously [9]. Briefly, HEK293 cells, maintained in Dulbecco's Modified Eagle Medium (DMEM) supplemented with 5% fetal bovine serum (FBS), 100 U/mL penicillin, and 100 µg/mL streptomycin, were incubated at 37 °C with 5% CO_2. The cells were co-transfected with pAAV2/8, pHelper, and the recombinant AAV transfer plasmid containing either GFP or the ApoE isoforms, using the calcium phosphate precipitation method. After three days of incubation at 37 °C, cells were harvested and lysed through three freeze-thaw cycles. The lysates were treated with 50 U/mL Benzonase to degrade nucleic acids, followed by iodixanol gradient centrifugation. The iodixanol gradient fractions were further purified using a HiTrap Q column (GE Healthcare) and concentrated with a Vivaspin concentrator (Bohemia, NY). The genomic titer of each virus was determined by quantitative PCR.

Intracranial injections in neonatal mice

Intracranial injection of AAV8 viruses in neonatal mice was performed as previously described [10]. In brief, postnatal day 2 (P2) mice were cryoanesthetized on ice for 5 min before receiving intracerebroventricular injections of 2 µl (~10^{13} viral particles/ml) into both hemispheres. Injections were performed using a Hamilton syringe (7642–01) fitted with a 30-gauge needle (7803–07, Hamilton). Following the procedure, pups were placed on a warming pad until they regained normal coloration and movement, after which they were returned to their mothers for recovery. In total, 20 mice received the APOE3 containing construct, 17 mice received APOE3Ch, and 12 mice received the *GFP* control construct. In the APOE3Ch group, 1 female mouse was prematurely sacrificed because of seizures; 2 male mice presented with hydrocephaly and had to be excluded.

TaqMan SNP genotyping assay (ThermoFisher assay ID: C__27540615_20) was used to detect the rs121918393 variant (*APOE3Ch*) and ensure correct group assignment, following the manufacturer's instructions. Genotype calls were determined on TaqMan Genotyper Software (ThermoFisher) (Supp. Figure 1A). To assess variability in AAV delivery within the APOE-expressing groups, RT-QPCR was performed on extracted genomic DNA using primers targeting the human *APOE* transgene (*APOE*, forward: 5′-CACTGGGTCGCTTTTGGGA-3′ and reverse: 5′-CAGTTGTTCCTCCAGTTCCGA-3′). The mouse *Actb* gene was used to normalize for genomic DNA input (*Actb*, forward: 5′-AGTGTGACGTTGACATCCGTA-3′ and reverse 5′-GCCAGAGCAGTAATCTCCTTC-3′). Standard curves were generated using serial dilutions of *APOE* and *Actb* plasmid DNA (Origene, Rockville, MD) to quantify absolute copy numbers. Relative vector genome abundance was compared across injection groups (Supp. Figure 1B). Overall, in the APOE3 group, one female had to be excluded due to absence of genomic APOE and APOE at the transcript level and one male had to be excluded due to mis-genotyping. In the APOE3Ch group, one female had to be excluded due to no available tissue for downstream experiments. Quantification of APOE expression in the cortex was performed via western blot and revealed that astrocyte-specific overexpression via AAV8-GFAP-APOE3 led to approximately 1.5-fold higher total APOE compared to the endogenous expression in an APOE3-targeted replacement mouse (Supp. Figure 1C).

Mouse brain tissue preparation and processing

Mice were sacrificed at two different timepoints post AAV virus injections, at 3.5 months (referred to as the early amyloid cohort), and at 8 months (late amyloid cohort). Mice under isoflurane anesthesia were perfused transcardially with phosphate-buffered saline (PBS), and their brains were swiftly harvested. The right hemibrains were fixed in 4% paraformaldehyde for immunofluorescence analysis, while the left hemibrains were dissected and stored at −80 °C for subsequent biochemical analysis. Cortical tissue was pulverized and split for protein extraction as well as genomic DNA/RNA isolation. Genomic DNA and RNA were isolated using the Qiagen AllPrep DNA/RNA kit as per the manufacturer's instructions. Total DNA and total RNA concentrations were determined by spectrophotometry using Nanodrop. Samples were stored at −80 °C until use. For protein extraction, soluble, detergent-soluble, and insoluble protein fractions were obtained from sequential extraction. Cortical tissues were homogenized with a Polytron homogenizer in ice cold Tris-buffered saline (TBS, pH 7.4) supplemented with protease inhibitor and phosphatase inhibitor cocktails (Roche). Homogenates were ultracentrifuged at 100,000 x*g* for 60 min at 4 °C and supernatants were collected (soluble fraction). Pellets were resuspended in TBS-X (TBS, 1% Triton X-100) supplemented with protease inhibitor and phosphatase inhibitor, sonicated, and ultracentrifuged as detailed above. Supernatants were collected (detergent soluble fraction), and the residual pellets were resuspended in 70% formic acid (FA), sonicated, and incubated overnight at 4 °C with end-over-end agitation. After ultracentrifugation as above, resulting supernatants were collected and neutralized with 20 volumes of 1 M Tris (pH 11) (insoluble fractions). All fractions were stored at −80 °C until further use.

Real-time quantitative PCR

Real-time quantitative PCR (RT-qPCR) reactions were run in duplicate on a QuantStudio 7 Flex Real-Time PCR System (Thermo Fisher Scientific). To assess changes in gene and transgene expression, extracted total RNA was first reverse transcribed using the iScript cDNA synthesis kit (Bio-rad). RT-qPCR was performed with SYBR Green Supermix (Bio-Rad). Relative gene expression was calculated using the $2^{-\Delta\Delta CT}$ method with *Actb* used as housekeeping gene. The following primers were used for cDNA amplification: *APOE*, 5′-CGTTGCTGGTCACATTCCT-3′ (forward), 5′-CTCAGTTCCTGGGTGACCTG-3′ (reverse); *Apoe*,5′-ATCTTGCGCAGGTGTGTGGAGA-3′ (forward), 5′-ACAGATCAGCTCGAGTAGCAAA-3′ (reverse) ; *Hspa5*, 5′- TTCAGCCAATTATCAGCAAACTCT-3′ (forward), 5′-TTTTCTGATGTATCCTCTTCACCAGT-3′ (reverse); *Hspb1*: 5′-GAGGAGCTCACAGTGAAGAC-3′ (forward), 5′-GCGTGTATTTCCGGGTGAAG-3′ (reverse); *Plin4*: 5′-GTGTCCACCAACTCACAGATG-3′ (forward), 5′-GGACCATTCCTTTTGCAGCAT-3′ (reverse); *Actb*, 5′-AGTGTGACGTTGACATCCGTA-3′- (forward), 5′-GCCAGAGCAGTAATCTCCTTC-3′ (reverse).

Western blotting

Soluble and detergent soluble fractions of cortical lysates were prepared in 5x Laemmli sample buffer (Genscript) containing β-mercaptoethanol (BME) and boiled for 5 min. Protein samples were resolved by SDS-PAGE using 4–20% precast TGX gels (Bio-rad) and transferred onto polyvinylidene difluoride (PVDF) membranes (Bio-rad). Membranes were blocked with 5% non-fat milk in PBS and incubated with primary antibody (Supp. Table 1) in 5% non-fat milk in PBS-T (1X PBS, 0.05% Tween-20) at 4 °C overnight. Membranes were washed with PBS-T and incubated with secondary antibody (Supp. Table 1) for 1 h at room temperature. For western blot analysis, Aβ42 samples were prepared in 4x LDS sample buffer (Thermo Fisher) without BME and were not boiled to preserve aggregates structures. Proteins were resolved by SDS-PAGE on 4–20% NuPAGE Bis-Tris gels (Thermo Fisher) using NuPAGE MES SDS Running buffer (Thermo Fisher). Following electrophoresis, proteins were transferred to PVDF membranes (Millipore) using NuPAGE transfer buffer (Thermo Fisher). Membranes were blocked with 5% non-fat milk in TBS-T (1XTris-buffered saline, 0.1% Tween-20) for 30 min at room temperature and incubated at 4 °C overnight with primary antibody diluted in blocking buffer. After washing with TBS-T, membranes were incubated with secondary antibody in blocking buffer for 1 h at room temperature. Membranes were either

directly imaged using the Odyssey imaging system or developed using ECL, SuperSensitive ECL, or Supersignal™ West Pico PLUS HRP substrate and signals detected using the Chemidoc Imaging system (Bio-Rad). Western blot bands were quantified using the ImageJ software.

Immunofluorescent staining

Following overnight fixation in 4% PFA, hemibrains were transferred to 30% sucrose for cryoprotection. Hemibrains were then embedded in Tissue-Tek® Optimal Cutting Temperature Compound (4583, Sakura) and frozen on dry ice. Frozen brains were sectioned coronally into 40 µm thick slices using a freezing-sliding microtome (LEICA SM2400, Leica). Free-floating brain slices were permeabilized with 0.2% Triton-X 100 in PBS for 30 min and blocked with 5% normal goat serum at room temperature. Antibodies for staining were diluted in antibody dilution buffer (1% normal goat serum, 0.1% Triton-X 100 in PBS). To stain for S100β, antibody dilution buffer was supplemented with 1 M glycine. Sections were incubated with primary antibodies (Supp. Table 2) overnight at 4°C, followed by incubation with fluorescently labelled secondary antibodies (Supp. Table 2) for 1.5 h at room temperature. DAPI (D1306, Invitrogen) was used as a counterstain.

For staining of fibrillar amyloid plaques with X-34 (SML1954, Sigma-Aldrich), brain slices were incubated with X-34 in 40% ethanol/PBS at room temperature for 20 min prior to immunostaining. Sections were mounted on glass slides with Vectashield prior to imaging. Images were obtained by fluorescence microscopy (BZ-X810, Keyence) or confocal laser scanning fluorescent microscopy (LSM510 Invert, Zeiss). All images were analyzed using Imaris 10.0.1 using the surfaces feature, including quantification of immunopositive sum volumes, sum intensities and mean intensities. Plaque sphericity was quantified using the "sphericity" parameter generated by Imaris, which computes a dimensionless value describing how closely the three-dimensional shape of each segmented volume approaches that of a perfect sphere. This morphological feature has been previously employed to characterize amyloid plaque structure in AD models [11]. Plaque-associated Iba1-positive sum volume and plaque-associated BACE1-positive sum volume were determined within a 10 µm radius. The spots feature was used to determine shortest distance of microglia to plaque surface within a 10 µm radius. Measurements were quantified within individual region of interest z-stacks, each corresponding to a single image field with identical dimensions. Data were averaged across 3–5 ROIs per mouse brain (5 ROIs in cortex and 3 ROIs in hippocampus; 3 ROIs in cortex with confocal imaging). All reported values reflect measurements within the imaged field only, with no extrapolation or normalization to total tissue volume.

For staining of iPSC-derived astrocytes, cells were fixed for 10 min with 4% PFA, permeabilized with 0.3% Triton-X 100 in PBS for 10 min, and blocked in blocking buffer (1% BSA in PBS, Milteny Biotec, 130-091-376) for 1 h. Cells were incubated with primary antibodies (Supp. Figure 2A) in blocking buffer overnight at 4 °C, followed by fluorescently labelled secondary antibodies (Supp. Figure 2B) for 1 h at room temperature. DAPI was used as counterstain.

Enzyme-linked immunosorbent assay (ELISA)

Levels of Aβ40 and Aβ42 in the soluble, detergent soluble and insoluble fractions of mouse cortical brain lysates were detected by in-house sandwich ELISA, as previously described. Briefly, Aβ40 and Aβ42 in lysates were captured onto a coated 96-well plate by end-specific monoclonal antibodies (mAb 13.1.1 and mAb 2.1.3 respectively). The human Aβ1-16–specific monoclonal antibody Ab5 (ab274112, Abcam) was conjugated to horseradish peroxidase (HRP) and used as detection antibody. Plates were developed with the addition of 3,3′,5,5′- tetramethylbenzidine (TMB) substrate, and the reaction was terminated with 1 M sulfuric acid. Absorbance was read at 450 nm using a Synergy HT microplate reader (BioTek Instruments Ink). Oligomeric Aβ (oAβ) levels were measured in the detergent soluble fractions using a commercially available kit (BEK-2215-1P, Biosensis). Target protein concentrations were calculated based on the corresponding standard curves. ELISA measurements were normalized to the samples' total protein concentration, which was determined using Pierce Detergent Compatible Bradford assay kit (ThermoFisher Scientific).

Bulk RNA-sequencing

The RNA integrity number (RIN) of extracted RNA was measured by the Agilent 2100 bioanalyzer to ensure adequate quality for bulk RNA-sequencing analysis. A total of 23 samples were sequenced. RNA libraries were prepared from 200 ng of total RNA using the TruSeq RNA Sample Prep Kit v2 (Illumina, San Diego, CA) according to the manufacturer's instructions and sequenced with Illumina HiSeq 4000 at the Mayo Clinic Center for Individualized Medicine's Medical Genome Facility. Reads were mapped to the mouse reference genome mm10, and raw gene read counts were generated using the Mayo Clinic RNA-sequencing analytic pipeline, MAP-RSeq version 3.0 (Kalari et al., 2014).

Gene normalization, differential gene expression and pathway analysis

Conditional quantile normalization (CQN) was applied on the raw gene counts to correct for GC bias and gene length differences and to obtain similar quantile-by-quantile distributions of gene expression levels across samples. Based on the bimodal distribution of CQN-normalized and log2-transformed reads per kilobase per million (RPKM) gene expression values, 14,504 genes had average log2 RPKM > 1 in either the APOE3 or APOE3-christchurch groups and were considered expressed and used for downstream analysis. Differential gene expression analyses were performed by Partek Genomics Suite (Partek Inc., St. Louis, MO) using CQN-normalized log2RPKM values. Gene expressions between APOE3Ch and APOE3 were compared using Analyses of Variance models (ANOVA) while adjusting for sex. The Benjamini-Hochberg procedure was applied to adjust for multiple testing. Differentially expressed genes (DEG) were defined by thresholds of BH-adjusted p value < 0.05 and |fold change| (FC) ≥ 1.2. Pathway analyses of differentially expressed genes were performed using Ingenuity Pathway Analysis (QIAGEN Inc., https://digitalinsights.qiagen.com/products/ingenuity-pathway-analysis) (Krämer et al., 2014).

Weighted gene co-expression network analysis

Weighted gene co-expression network analysis (WGCNA) was performed using CQN-normalized and $\log_2$-transformed RPKM values. Based on the relationship between power and scale independence, the power of 8 was chosen to build scale-free topology using signed hybrid network. We set the minimum modules size as 60, and merged modules whose correlation coefficients were greater than 0.7 (mergeCutHeight = 0.3). Each module was summarized by the first principal component of the scaled module expression profiles, termed module eigengene (ME). For each module, the module membership (MM) was defined as the correlation between gene expression values and ME. Intramodular hub genes are genes with the highest connectivity to other genes within a given module, and were selected based on the p values of MM. To assess the correlation of modules to *APOE* genotype and sex, we defined the APOE3Ch as 1 and APOE3 as 0; female as 1 and male as 0. Modules were annotated using R package anRichment.

Astrocyte differentiation and conditioning

Human iPSC-derived astrocytes from the KOLK2.1J line and isogenic *APOE3Ch* line were generated as described, with modifications [12]. iPSCs were first differentiated into neural stem cells using a commercial neural induction medium (StemCell Technologies) following manufacturer's protocol. Briefly, iPSCs were dissociated into a single-cell suspension and into Aggrewell 800 plates (StemCell Technologies) in neural induction medium supplemented with 10 µm Y-27632 (ROCK inhibitor) to promote the formation of uniformly sized embryoid bodies (EBs). EBs were cultured for 5–7 days with daily medium changes and were then collected and transferred onto Matrigel-coated dishes in fresh neural induction medium for an additional 5–7 days to induce neural rosette formation. Neural rosettes were manually selected and re-plated onto Matrigel-coated dishes in neural induction medium to allow further expansion. For neural progenitor cell (NPC) specification, rosettes were dissociated into single-cells and seeded onto a Matrigel-coated dish neural progenitor cell medium (StemCell Technologies) with ROCK inhibitor and cultured for additional 10–14 days. NPCs were expanded and cryopreserved for downstream differentiation into astrocytes. To generate iPSC-derived astrocytes, NPCs were plated onto Matrigel-coated plates in Astrocyte Medium, supplemented with 1% Pen-Strep, 1% Astrocyte Growth Serum, and 2% Fetal bovine serum (kit from ScienCell). Cells were passaged as needed upon reaching 80% confluency. iPSC-derived astrocytes were matured for 35-45 days following differentiation from NPCs. Cells were incubated in Astrocyte medium without FBS for 3 days to generate conditioned medium used in downstream applications.

Recombinant Aβ42 aggregation assay

Recombinant Aβ42 monomer (1 mg/ml) (Stressmarq, #SPR-485) was incubated at room temperature for 10 min. To dissolve the peptide,

7.5 µl of cold DMSO (Sigma Aldrich) was added, followed by 92.5 µl of cold PBS (PH 7.4) to reach a final concentration of 220 µM monomeric Aβ42. The solution was centrifuged at 14,000 xg for 10 min at 4 °C to remove insoluble aggregates. The supernatant was carefully transferred to a new tube, and 1400 µl of cold PBS was added to adjust the final concentration to 14 µM. Aβ42 was then mixed at a 1:1 ratio with serum-free conditioned medium from APOE-expressing astrocytes, which had been dialyzed using spin desalting columns (Thermo Fisher) according to the manufacturer's protocol. Dialyzed astrocyte culture media without FBS was used as control. The Aβ42 mixtures in media were incubated at 37 °C with 200 RPM shaking for 3 days.

Statistical analyses

All data were analyzed by two-way analysis of variance (ANOVA) to adjust for sex differences given prior reports of sex-dependent effects on pathology in the 5xFAD model [13], with a Tukey's post-hoc test using GraphPad Prism 10. Raw values and full two-way ANOVA outputs, including sex as a factor, are provided in Supplemental Data File 1. qPCR analysis was performed on log2 transformed fold change. Data were presented as mean ± SEM; closed circles were used for male datapoints, and open circles for female datapoints. A P value of < 0.05 was considered statistically significant.

RESULTS

AAV-based astrocytic expression of APOE3 and APOE3-Ch in 5xFAD mice

To investigate whether astrocytic expression of APOE3Ch mitigates amyloid pathology in vivo, we injected AAV8-GFAP vectors expressing APOE3, APOE3Ch, or GFP intracranially in the brains of 5xFAD mice at postnatal day 2 (P2). The mice were analyzed at 3.5 months of age (early amyloid) and 8 months of age (late amyloid), at which time brains were collected for biochemical and histological analyses (Fig. 1A). Representative GFP images demonstrate the distribution of AAV-GFAP-driven expression in cortex (Supp. Figure 1D), consistent with prior reports [10]. Viral expression was astrocyte-specific, with strong co-localization of GFP in GFAP-positive and S100b-positive astrocytes but not in NeuN-positive neurons or IBA1-positive microglia (Fig. 1B). Since the expression was robust in the cortex, we next quantified *APOE* expression in cortical tissue. No significant differences in *APOE* DNA were detected between the APOE3 and APOE3Ch groups, ensuring that differences between the two APOE groups were not due to differences in transduction efficiency (Supp. Figure 1B). Similarly, the mRNA levels of *APOE* in the APOE3 and APOE3Ch groups were comparable across groups, with no significant differences between APOE3 and APOE3Ch. GFP controls lacked human *APOE* mRNA expression (Fig. 1C). Expression of *APOE* did not result in significant changes in mouse endogenous *Apoe* transcripts compared to GFP (Fig. 1C). Biochemical analyses of cortical lysate fractions revealed that APOE3Ch expressing mice exhibited higher overall human APOE (hAPOE) levels compared to APOE3, particularly in the soluble (Fig. 1D, E) and detergent-soluble fractions (Fig. 1F, G) while insoluble pools remained unchanged (Supp. Figure 1E). These findings were also observed in the 3.5-month-old cohort, confirming that the higher levels of APOE3Ch are a consistent feature across ages (Supp. Figure 2A, B, D, E). In addition, soluble protein levels of mouse APOE (mAPOE) were significantly reduced in APOE3 and APOE3Ch groups compared to GFP at 8 months of age, but not 3.5 months of age (Fig. 1D, E, and Supp. Figure 2A–C). Detergent-soluble mAPOE levels remained unchanged in both early and late amyloid stages (Fig. 1F, G, and Supp. Figure 2D–F). Taken together, these data demonstrate that astrocytic expression of APOE3Ch leads to increased cortical accumulation compared to APOE3, which is not due to differences in viral delivery or transcriptional regulation. The increased detergent solubility of APOE3Ch further suggests potential differences in its biochemical properties. Additionally, soluble mAPOE protein levels were significantly reduced at 8 months of age in both APOE3 and APOE3Ch groups, despite unchanged mRNA levels. The observation suggests that the effect of hAPOE on mAPOE levels may reflect indirect consequences of amyloid pathology context. It is also possible that prolonged hAPOE expression alters mAPOE protein handling that emerges only at later stages.

Astrocytic APOE3 and APOE3Ch expressions ameliorate amyloid pathology and plaque-associated dystrophic neurites in 5xFAD mice

To investigate the potentially different effect of APOE3 and APOE3Ch on Aβ solubility, the mouse cortical samples were sequentially homogenized through TBS, TBSX, and FA buffers. Soluble Aβ40 and Aβ42 levels in TBS cortical lysates of 8-month-old 5xFAD mice were not significantly different across groups (Fig. 2A). Detergent-soluble Aβ40 in TBSX fraction was significantly reduced with astrocytic APOE3Ch expression compared to GFP, while Aβ42 levels were unchanged (Fig. 2B). These findings suggest that APOE3Ch has modest effects on early, soluble pools of Aβ. Insoluble Aβ40 levels in the FA fraction was decreased with astrocytic APOE3 and APOE3Ch relative to GFP controls, whereas insoluble Aβ42 levels were unaffected (Fig. 2C). Oligomeric Aβ (oAβ) levels in the detergent-soluble pool were measured to further assess the aggregation state of Aβ. The APOE3 group exhibited significantly higher levels of oAβ compared to GFP group, while oAβ levels in the APOE3Ch group resembled the GFP group (Fig. 2D). Thus, while both astrocytic expression of APOE3 or APOE3Ch reduced insoluble Aβ40 levels, only APOE3Ch significantly lowered detergent-soluble Aβ40 compared to GFP. APOE3Ch also showed a modest effect on increasing oAβ levels compared to APOE3. These results suggest a distinct aggregation-modifying effect of APOE3Ch. APP expression was unchanged across groups in the 8-month cohort (Fig. 2E, F), indicating that astrocytic APOE expression does not alter APP transcription or steady-state levels. No significant changes in insoluble Aβ levels (Supp. Figure 2G, H) or differences in detergent-soluble oAβ (Supp. Figure 2I) were observed across groups in the early amyloid 3.5-month-old cohort. Subsequent histological analysis of amyloid pathology and related toxicity therefore focused on the 8-month timepoint.

Histological characterization of amyloid plaque burden using MOAB2 antibody to detect total Aβ species and X-34 to stain amyloid cores confirmed that both APOE3 and APOE3Ch astrocytic expression resulted in significant reduction of Aβ deposition in the cortex compared to GFP (Fig. 3A–C). In the hippocampus, fibrillar amyloid burden was also significantly lower in the APOE groups compared to GFP, while no differences were detected in total Aβ deposition (Fig. 3A, D, and E). We then measured total BACE1 immunoreactivity across the ROI. Plaque-associated BACE1, defined as signal within 10 µm of X-34+ plaques, was also quantified to specifically capture local amyloid-driven dystrophic neurites. Dystrophic neurites represented by plaque-associated BACE1 in the cortex were significantly reduced in the APOE3Ch group compared to GFP, while mice in the APOE3 group showed no significant reduction. This quantification revealed similar reductions with astrocytic APOE expression relative to GFP, with no significant difference between APOE3 and APOE3Ch (Fig. 3F–J). These results indicate that astrocytic APOE expression reduces neuritic damage in the vicinity of fibrillar amyloid plaques, with APOE3Ch showing modestly superior effects. While protective effects of APOE3Ch on tau pathology are implicated [3], we did not detect evident effects of astrocytic expressions of APOE3 and APOE3Ch on endogenous mouse tau accumulation and phosphorylation in the mouse brains (Supp. Figure 3).

Astrocytic APOE3 and APOE3Ch expressions modulate glial response to amyloid pathology in 5xFAD mice

To evaluate the impact of APOE3 and APOE3Ch expression on glial reactivity, we first examined APOE levels and distribution in the

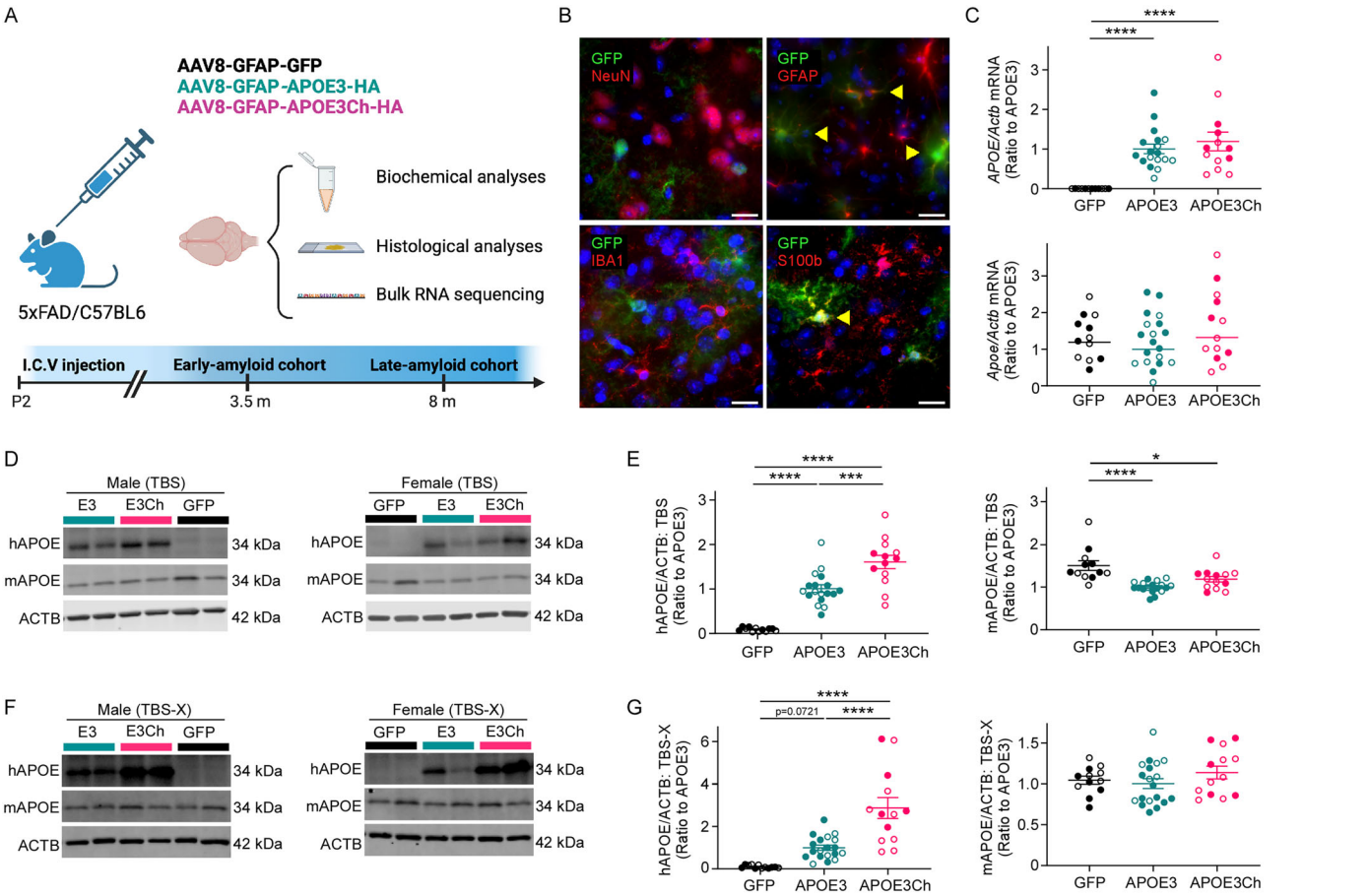


Fig. 1 Model overview and characterization of astrocytic APOE3Ch expression in 5xFAD mice. (**A**) Schematic experimental procedures of AAV-based astrocytic expression of GFP, APOE3, and APOE3Ch in 5xFAD mice are shown. Created with BioRender.com (**B**) Representative images of mouse cortex stained for GFP, NeuN, GFAP, Iba1 and S100b from the AAV8-GFAP-GFP injection group at 8 months of age are shown. Scale bar: 20 µm. (**C**) Human *APOE* and endogenous mouse *Apoe* mRNA levels in the mouse cortex were measured by RT-qPCR at 8 months of age, normalized to those of β-actin (*Actb*). (**D, E**) Amounts of human APOE (hAPOE) and mouse APOE (mAPOE) in the soluble TBS fractions of mouse cortical lysates were measured by western blot at 8 months of age, normalized to those of β-actin (ACTB). (**F, G**) Amounts of hAPOE and mAPOE in the detergent soluble TBS-X fractions of mouse cortical lysates were measured by western blot at 8 months of age, normalized to those of ACTB. Data are expressed as means ± SEM (GFP: N = 12; APOE3: N = 18; APOE3Ch: N = 13; Open circles: females, closed circles: males). Injection group differences were analyzed using two-way ANOVA, with Tukey's multiple comparisons test, adjusting for sex. *, $P < 0.05$; ***, $P < 0.001$; ****, $P < 0.0001$.

brains of 5xFAD mice at 8 months of age. As the AAV construct codes APOE tagged with HA in the C-terminus, we found that most of the immunostaining of APOE was co-localized with HA (Supp. Figure 4A), consistent with prior reports using this construct [10]. HA immunostaining was partially overlapped with X-34+ plaque cores in both APOE3 and APOE3Ch groups, with no evident difference between the two groups. Despite higher soluble human APOE level in the cortical lysates in the APOE3Ch group compared to APOE3 group (Fig. 1D, E), there was no significant difference in the cortical and hippocampal positivity of HA immunostaining signal between the groups (Fig. 4A, B, Supp. Figure 4B, C). These findings suggest the soluble abundance of APOE3Ch and APOE3 does not affect their tissue accumulation or aggregation profiles (Supp. Figure 1D).

We next assessed the effects on glial activations by quantifying brain GFAP and IBA1 immunoreactivity. There was a decrease in cortical GFAP volume that was statistically significant with APOE3 and trending down with APOE3Ch astrocytic overexpression, compared to GFP (Fig. 4A, C). There was no significant difference in hippocampal GFAP positivity among the groups (Supp. Figure 4B, D). In addition, astrocytic APOE expression affected microglial activation, with total cortical IBA1 volume significantly reduced in APOE3 and APOE3Ch groups compared to GFP controls (Fig. 4A, D), without significant differences between the APOE groups. While CD68 positivity showed a trend toward reduction with astrocytic APOE3Ch in the cortex, no significant differences were observed between the APOE groups (Fig. 4A, E). No differences in hippocampal IBA1 and CD68 positive volumes were observed between the GFP and APOE groups (Supp. Figure 4B, E, F). APOE3Ch resulted in a trending increase in mean X-34 intensity within plaques compared to GFP, with no difference to APOE3 (Fig. 4F, J). Because of these structural differences in plaque morphology, we also evaluated the spatial organization of microglia around plaques. We quantified the proportion of IBA1 signal within 0–10 µm of X-34-positive plaques to define direct microglial-plaque interaction. Plaque-associated IBA1 signal, expressed as a percentage of total IBA1 volume, was significantly lessened with APOE3 and APOE3Ch astrocytic expression compared to GFP. No difference was observed between APOE3 and APOE3Ch groups (Fig. 4G). Both APOE3 and APOE3Ch astrocytic expression increased the average distance of plaque-associated microglia to plaques within 10 µm, indicating reduced physical engagement of microglia with plaques (Fig. 4H). A closer examination of X-34 stained fibrillar

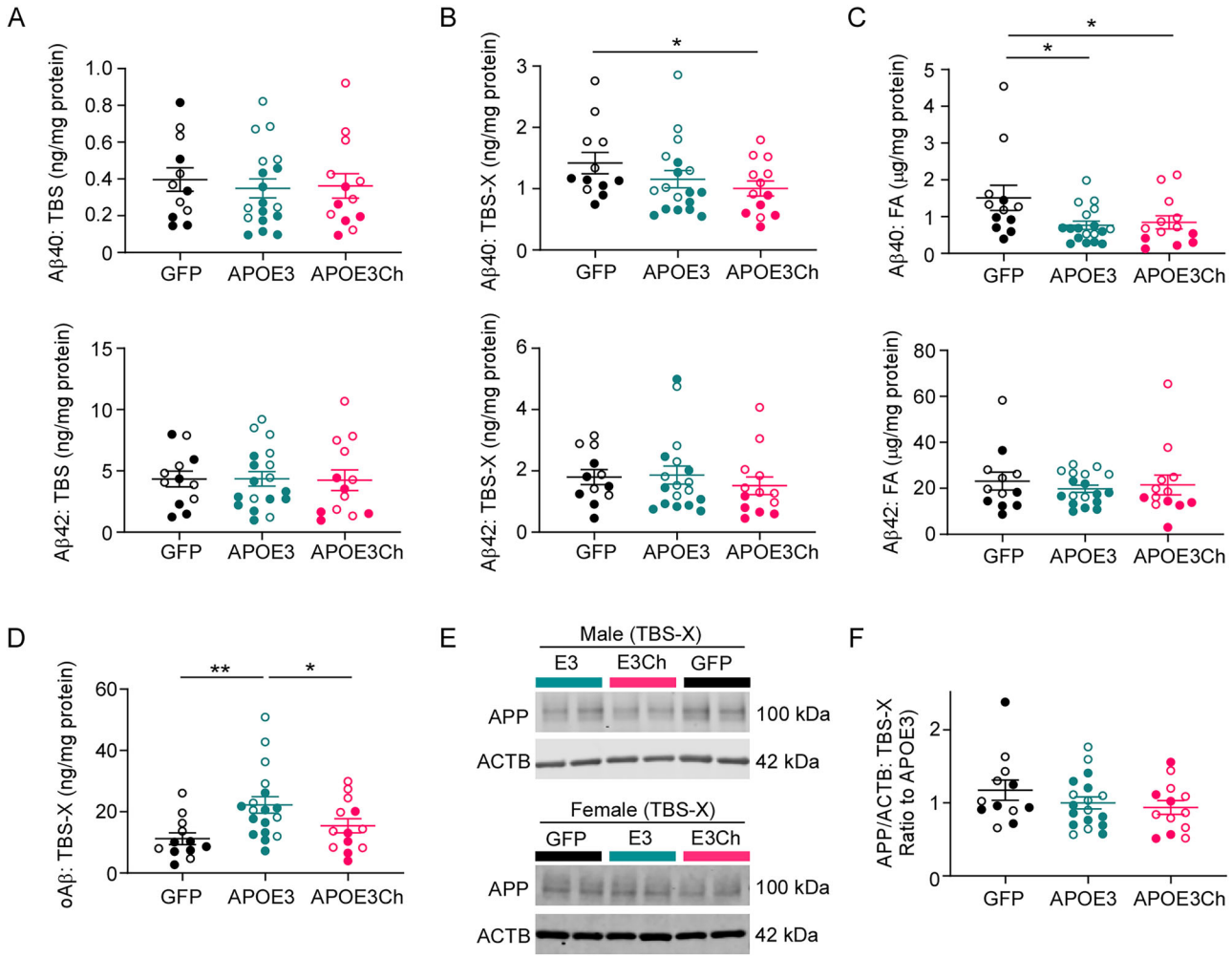


Fig. 2 Impacts of APOE3Ch astrocytic expression on brain Aβ accumulation in 5xFAD mice. (**A**-**C**) Aβ40 and Aβ42 levels in the soluble TBS fractions (**A**), detergent soluble (TBS-X) fractions (**B**), and insoluble neutralized formic acid (FA) fractions (**C**) of cortical brain homogenates in the 5xFAD mice were measured by ELISA at 8 months of age, normalized to protein concentrations of each fraction. (**D**) Soluble oligomeric Aβ (oAβ) levels in the TBS-X fractions of cortical brain homogenates in the 5xFAD mice were measured by ELISA at 8 months of age, normalized to protein concentrations. (**E, F**) Amounts of full-length APP in the TBS-X fractions of mouse cortical lysates were measured by western blot at 8 months of age, normalized to those of β-actin (ACTB). Data are expressed as means ± SEM (GFP: N = 12; APOE3: N = 18; APOE3Ch: N = 13; Open circles: females, closed circles: males). Injection group differences were analyzed using two-way ANOVA, with Tukey's multiple comparisons test, adjusting for sex. *, $P < 0.05$; **, $P < 0.01$.

plaque highlighted structural differences, as measured by plaque sphericity [14, 15] and mean fluorescent intensity of X-34 which was used as a proxy for β-sheet density within plaques. Astrocytic APOE overexpression resulted in significantly more spherical fibrillar Aβ plaques than GFP. This effect was more pronounced with APOE3Ch, which led to a significant increase in average plaque sphericity than APOE3 (Fig. 4I, J). Together, these results indicate that astrocytic APOE3 and APOE3Ch expressions reduce global gliosis, in particular microgliosis, and alter microglial plaque association particularly in the cortex. Astrocytic APOE3Ch expression shows specific increased plaque sphericity and β-sheet density, consistent with a more compact and structurally organized plaque morphology.

APOE3Ch astrocytic expression alters brain transcriptional pathways related to protein folding, immune signaling, and nuclear regulation in 5xFAD mice

Bulk RNA-sequencing of cortical tissue from 8-month-old male 5xFAD mice expressing astrocytic APOE3 or APOE3Ch was performed to identify potential molecular differences between the two groups.

We found a limited set of differentially expressed genes (DEGs) between groups (Fig. 5A). *Hspa5* coding a key endoplasmic reticulum (ER) stress chaperone binding immunoglobulin protein (BiP) or GRP78, *Hspb1* coding heat shock protein family B member 1 (HSP-27), and *Plin4* coding perilipin 4 (PLIN4) were downregulated with astrocytic APOE3Ch compared to APOE3. Ingenuity pathway analysis (IPA) identified the unfolded protein response (UPR) as the top downregulated pathway in male mice of the APOE3Ch group. Other significantly downregulated pathways include ER stress and immunogenic cell death signaling pathways (Fig. 5B), supporting broader suppression of stress-related transcriptional programs with astrocytic overexpression of APOE3Ch compared with APOE3. RT-qPCR confirmed the significant reduction in *Hspa5* in both APOE3Ch and GFP groups compared to APOE3, while *Hspb1* was selectively downregulated in the APOE3Ch group. We did not detect significant downregulation of *Plin4* with APOE3Ch (Fig. 5C–E). In addition, weighed gene co-expression network analysis (WGCNA) was performed to further explore gene network dynamics. Pink module was upregulated with astrocytic APOE3Ch expression, associated with chromatin organization and transcriptional regulation, suggesting a shift toward nuclear activity in the context of reduced cellular

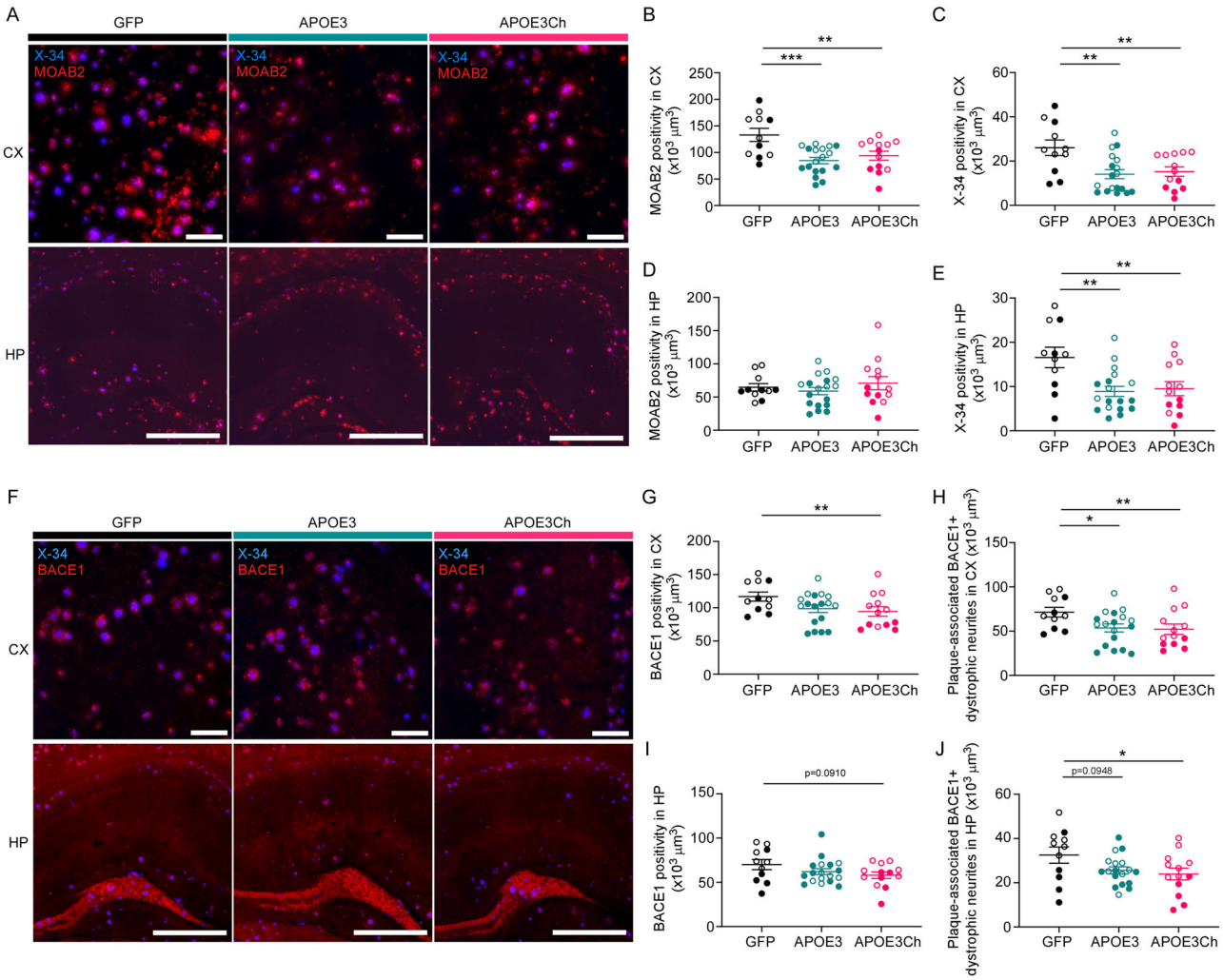


Fig. 3 Impacts of APOE3Ch astrocytic expression on amyloid pathology and dystrophic neurites in 5xFAD mice. (**A**) Representative images of amyloid pathology in 8-month-old 5xFAD mouse brains are shown. The cortex (CX) and hippocampus (HP) were stained with X-34 (blue, fibrillar plaques) and MOAB2 (red, pan-Aβ antibody). Scale bar: 100 µm (CX), 500 µm (HP). (**B**-**E**) Amyloid burdens for MOAB2-positive volumes and X-34-positive volumes in the CX (**B**, **C**) and HP (**D**, **E**) were quantified using Imaris. (**F**) Representative images of dystrophic neurites (BACE1, red) co-stained with fibrillar amyloid plaques (X-34, blue) in the CX and HP of 8-month-old 5xFAD mouse brains are shown. Scale bar: 100 µm (CX), 500 µm (HP). (**G**-**J**) Total BACE1-positive volumes and plaque-associated BACE1 within 10 µm of X-34+ plaques in the CX (**G**, **H**) and HP (**I**, **J**) were quantified using Imaris. Data are expressed as means ± SEM (GFP: N = 11; APOE3: N = 18; APOE3Ch: N = 13; Open circles: females, closed circles: males). Injection group differences were analyzed using two-way ANOVA, with Tukey's multiple comparisons test, adjusting for sex. *, P < 0.05; **, P < 0.01; ***, P < 0.001.

stress (Fig. 5F). Two modules (Brown and Red modules) were downregulated with astrocytic APOE3Ch expression: Brown module enriched for genes involved in immune signaling and membrane-associated activation (Fig. 5G), and Red module included genes involved in protein folding and proteasome function (Fig. 5H). While we did not detect evident changes in the levels of BiP, we observed the decrease at the protein level of HSP27. No significant difference was observed between APOE3 and APOE3Ch (Fig. 5I, J). PLIN4 immunoreactivity was primarily detected in SOX10-positive oligodendrocytes with signal appearing both in discrete puncta consistent with lipid droplets and in a more diffuse or perinuclear distribution (Fig. 5K). PLIN4 levels were particularly reduced with APOE3Ch expression compared to GFP (Fig. 5L, M). Together, these results demonstrate a reduction in the protein levels of PLIN4 and HSP27 with astrocytic expression of APOE3Ch compared with GFP. Although the protein changes do not replicate the RNA-seq finding directly, they provide complementary evidence that stress-response pathways are altered in APOE-expressing brains.

Human iPSC-astrocyte derived APOE3Ch reduces Aβ42 oligomer formation in vitro

One key difference between APOE3 and APOE3Ch astrocytic expression in 5xFAD mice was elevated APOE at the protein level in APOE3Ch. In turn, we observed subtle differences in Aβ processing and in plaque morphology between the APOE3 and APOE3Ch groups. To validate these differences, we generated iPSC-derived astrocytes expressing APOE3 or APOE3Ch (Fig. 6A) and assessed protein expression as well as Aβ-related phenotypes in vitro. Consistent with our in vivo data, APOE levels in astrocyte-secreted medium were higher in APOE3Ch astrocytes compared to APOE3, with no significant difference in mRNA levels (Fig. 6B–D). We then exposed astrocyte-conditioned media (ACM) to synthetic Aβ42 and monitored aggregation dynamics. Similar to what we observed in vitro, ACM from APOE3Ch astrocytes attenuated the levels of Aβ42 oligomers, suggesting changes in the APOE3Ch astrocyte secretome or APOE-dependent modulation of Aβ42 assembly (Fig. 6E–H).

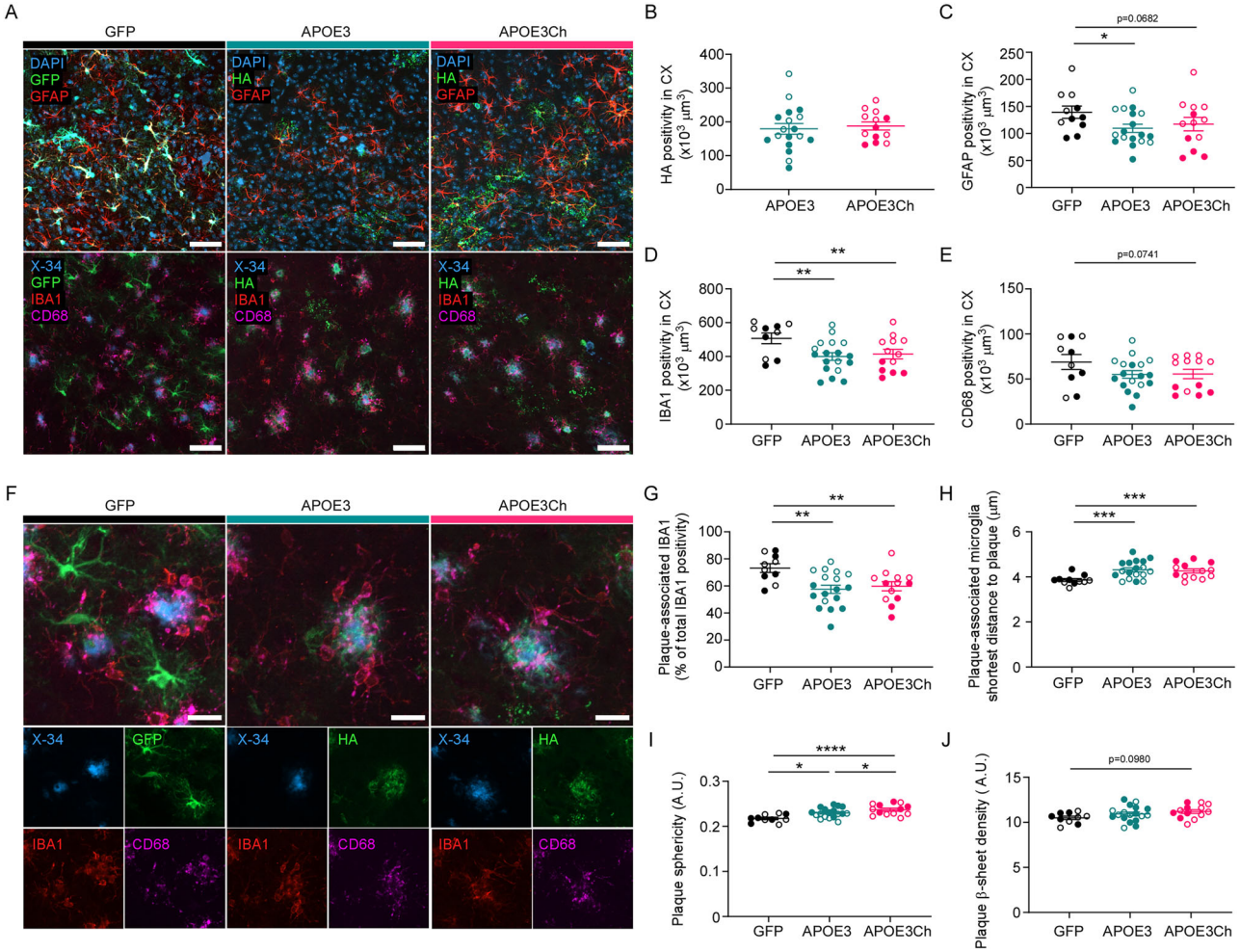


Fig. 4 Impacts of astrocytic APOE3Ch expression on amyloid-related glial activation in 5xFAD mice. (**A**) Representative images of astrocytes (top) and microglia (bottom) in 8-month-old 5xFAD mouse cortical sections are shown. Top: GFAP (red) for astrocytes, GFP or HA for APOE (green), and DAPI (blue) for nuclei were stained. Bottom: IBA1 (red) for microglia, CD68 (magenta) for activated microglia, GFP or HA for APOE (green), and X-34 (blue, fibrillar plaques) were stained. Scale bars: 50 µm. (**B, C**) APOE C-terminus HA positive-volumes (**B**) and GFAP-positive volumes (**C**) in the cortex (CX) were quantified using Imaris. (**D, E**) IBA1-positive volumes (**D**) and CD68-positive volumes (**E**) in the CX were quantified using Imaris. (**F**) Representative images of plaque-associated microglia in 8-month-old 5xFAD mouse cortical sections are shown. IBA1 (red) for microglia, CD68 (magenta) for activated microglia, GFP (green) for APOE, and X-34 (blue) for fibrillar amyloid plaques were stained. Scale bars: 20 µm. (**G, H**) Plaque-associated IBA1-positive microglia within 10 µm of a plaque (**G**) and the shortest distance from each plaque (**H**) in the CX were quantified using Imaris. (**I, J**) X-34-positive plaque sphericity (**I**) and the β-sheet density (**J**) in the CX were quantified using Imaris. Data are presented as means ± SEM (GFP: N = 11; APOE3: N = 18; APOE3Ch: N = 13; Open circles: females, closed circles: males). Group differences were analyzed using two-way ANOVA with Tukey's multiple comparisons test, adjusting for sex. *, P < 0.05; **, P < 0.01; ***, P < 0.001; ****, P < 0.0001.

DISCUSSION

The *APOE3Ch* variant has garnered significant attention when homozygosity for this rare *APOE3* variant was found to be resilient against the onset of a very aggressive form of autosomal AD case driven by the *PSEN1* p.E280A mutation, despite extensive amyloid deposition [3]. However, the extent to which these protective effects are cell-autonomous or require systemic APOE3Ch expression remains unknown. Therefore, in this study, we focused on the therapeutic potential of astrocytic APOE3Ch expression to modulate amyloid pathology in vivo. To model a therapeutically relevant scenario, we expressed human APOE3Ch in astrocytes of 5xFAD mice through intracerebroventricular injection of GFAP promoter-driven AAV into the neonatal pups. Our AAV-GFAP-APOE-HA construct has been previously validated [10] to produce correctly sized, secreted, and lipidated APOE, supporting its functional integrity in vivo. This was in the context of intact endogenous mouse APOE expression, allowing us to assess the effects of increasing a protective APOE3Ch without removing endogenous APOE function. By targeting astrocytes, the primary source of APOE in the CNS [8], we aimed to determine how enhancing APOE3Ch levels in this cell population could influence amyloid deposition under physiologically relevant conditions. Our strategy offers a translational framework to evaluate the therapeutic relevance of astrocyte-derived APOE3Ch in modulating amyloid pathology.

Astrocytic *APOE3Ch* gene delivery led to an increase in APOE protein level across soluble and detergent soluble cortical fractions compared to that through *APOE3* gene delivery, despite comparable mRNA levels between them. These observations suggest that APOE3Ch has better post-transcriptional stabilization or slower turnover and clearance than APOE3. This could translate into differences in lipidation status between the two isoforms. These observations could also reflect reduced cellular uptake and lysosomal degradation, consistent impaired receptor binding of APOE3Ch to its receptors such as LDLR and HSPGs [3, 16, 17], potentially resulting in diminished cellular uptake and clearance.

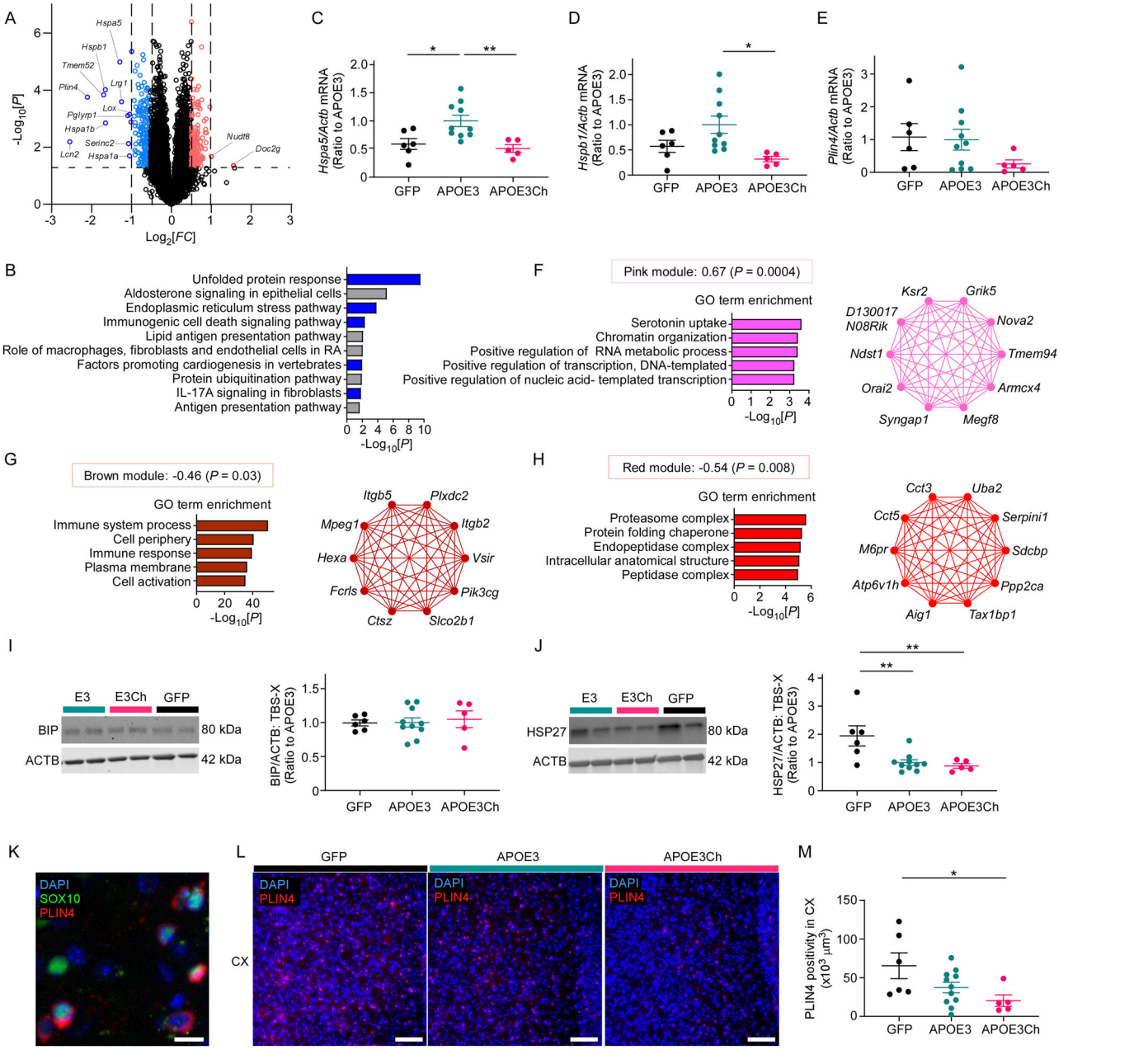


Fig. 5 Reduced inflammatory, protein degradation, and cellular stress pathways with APOE3Ch astrocytic expression in 5xFAD mice. (**A, B**) Bulk RNA-sequencing was performed in in the cortex of 8 months old male 5xFAD mice (APOE3: N = 6; APOE3Ch: N = 5). Volcano plot (**A**) affected by APOE3Ch astrocytic expression compared to APOE3 and Ingenuity Pathway Analysis of the DEGs (**B**) are shown. Pathway coloring follows IPA conventions: blue = predicted inhibition; grey = no predicted activation state. (**C-E**) The mRNA levels of selected DEGs, *Hspa5* (**C**), *Hspb1* (**D**), and *Plin4* (**E**) in the cortex of 8 months old male 5xFAD mice were measured by RT-qPCR in the mice, normalized to those of β-actin (*Actb*). (**F-H**) WGCNA in the RNA-seq identified one upregulated module (**F**, Pink) and two down-regulated modules (**G**, Brown; **H**, Red). The top 10 hub genes and top 5 GO terms of each module are shown. (**I, J**) Amounts of BIP (**I**) and HSP27 (**J**) in the TBS fractions of cortical lysates of 8 months old male 5xFAD mice were measured by western blot at 8 months old, normalized to those of β-actin (ACTB). (**K**) Immunostaining of PLIN4 (red) in the cortex of 5xFAD mice at 8-month of age shows localization primarily in oligodendrocytes, which were stained with oligodendrocyte marker SOX10 (green). DAPI was used as counterstain. Scale bar: 10 µm. (**L**) Representative immunostaining images of PLIN4 (red), oligodendrocyte marker SOX10 (green), and DAPI (blue) in the cortex (CX) of 8-month-old 5xFAD male mice are shown. Scale bar: 100 µm. (**M**) PLIN4-positive volumes in the CX were quantified using Imaris. Data are presented as means ± SEM (GFP: N = 6; APOE3: N = 10; APOE3Ch: N = 5). Group differences were analyzed using one-way ANOVA with Tukey's multiple comparisons test. *, P < 0.05; **, P < 0.01.

Interestingly, levels of insoluble APOE3Ch detected in the insoluble fraction were comparable to APOE3, suggesting that a smaller proportion of total APOE3Ch is found in insoluble aggregates compared to APOE3. This is of importance as recent evidence suggest that APOE aggregation in microglia initiates Aβ seeding in 5xFAD mice [18]. Although further studies are needed, the different biological properties of APOE3Ch may contribute to its protective effects against AD. In line with other studies [4, 7], we also found that astrocytic APOE3Ch expression reduces amyloid plaques in the 5xFAD mouse model. This is seemingly in contradiction with reports of astrocyte-specific deletion of APOE reducing amyloid pathology and associated toxicity [19]. However,

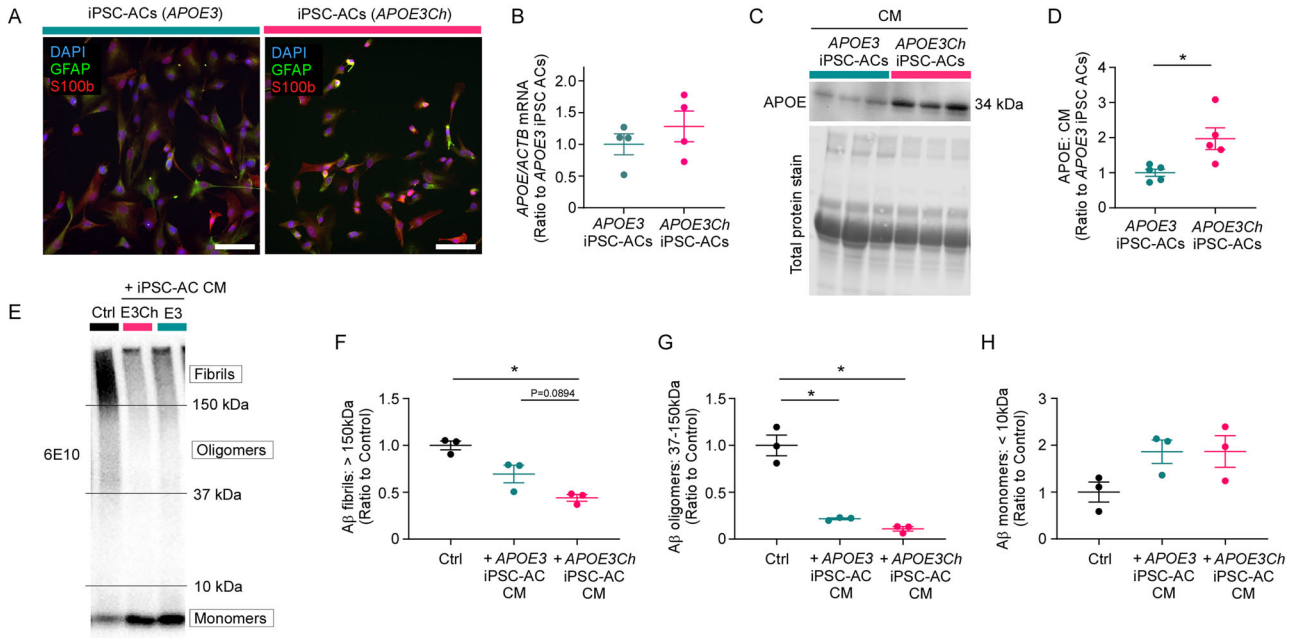


Fig. 6 Condition medium of iPSC-derived astrocytes with APOE3Ch inhibits Aβ oligomer formation in vitro. (**A**) Isogenic iPSC-derived astrocytes (iPSC-ACs) with homozygous *APOE3* (left) and *APOE3Ch* (right) were immunostained for GFAP (green) and S100b (red). DAPI (blue) stains nuclei. Scale bars: 100 µm. (**B**) *APOE* mRNA levels in the iPSC-ACs were measured by RT-qPCR, normalized to those of β-actin (*ACTB*). (**C, D**) Amounts of APOE in the conditioned medium (CM) from iPSC-ACs were measured by western blot, normalized to protein amounts of cell lysates. (**E**) Effects of the CM from iPSC-ACs on Aβ42 aggregation was assessed by western blot using 6E10 antibody. (**F-G**) Populations of Aβ fibrils (F; > 150 kDa), oligomers (G; 37–150 kDa), and monomers (**H**; < 10 kDa) were quantified. Data expressed as means ± SEM (n = 3–5 independent differentiation batches). Group differences were analyzed using two-tailed student t-test or one-way ANOVA with Tukey's multiple comparisons test. *, P < 0.05.

this discrepancy likely reflects difference in model context. In our system, human *APOE* is expressed in astrocytes on the background of endogenous mouse *Apoe*. These findings suggest that the effects of APOE are not dependent on just its presence or absence, but also on its levels and the cellular context.

In addition to resulting in an overall reduction in amyloid pathology, including decreased MOAB2 and X-34 plaque burden, APOE3 and APOE3Ch astrocytic overexpression resulted in a selective reduction in insoluble $A\beta_{40}$. While this indicates a broad decrease in parenchymal Aβ deposition, the specific decline of $A\beta_{40}$, the predominant species deposited in cerebral vessel walls [20], raises the possibility of effects on vascular amyloid. However, the 5xFAD model does not robustly develop cerebral amyloid angiopathy (CAA) under standard conditions [21], therefore we did not assess CAA directly. While both astrocytic APOE3 and APOE3Ch expressions reduced insoluble $A\beta_{40}$ and amyloid burden compared to GFP controls, APOE3Ch selectively lowered detergent-soluble levels of $A\beta_{40}$ compared to GFP controls, and oligomeric Aβ levels compared with APOE3, suggesting the distinct biochemical interactions influencing Aβ deposition or clearance [2, 3, 22, 23]. This would align with previous in vitro studies reporting decreased Aβ aggregation in the presence of recombinant APOE3Ch compared with APOE3 [3, 24]. APOE3 appears to promote oligomer-prone aggregation of Aβ compared to APOE3Ch, even while reducing total amyloid deposition as shown in both our in vivo and in vitro models. Evidence suggests that Aβ oligomers are the most neurotoxic Aβ species, driving synaptic dysfunction, neuronal loss and cognitive decline in AD [25–27]. Reduced oligomeric Aβ levels with astrocytic APOE3Ch expression may contribute to a less toxic environment, which would be consistent with our bulk RNA sequencing dataset. For instance, we found that astrocytic APOE3Ch expression leads to suppression of immune-related gene networks in the brains of male 5xFAD mice, suggesting the ameliorated Aβ-induced neuroinflammation. As the downregulated DEGs include *Hspa5*, *Hspb1*, *Hspa1a*, and *Hspa1b*, unfolded protein response and ER stress pathway are identified as pathways negatively affected by APOE3Ch compared to APOE3. Since AD pathology induces ER stress and excessive UPR activation [28], ER chaperones and small heat-shock proteins likely increase in neurons and glia in AD at the late stage [29, 30]. Thus, the observed gene reduction by APOE3Ch may reflect a diminished demand for chaperone-mediated stress response due to reduced misfolded protein stress in the context of dampened amyloid pathology. Although further studies are needed, protective roles of APOE3Ch are possibly mediated by its superior functions to prevent protein misfolding including Aβ.

We also observed reduction in the levels of PLIN4 with APOE3Ch compared to GFP. PLIN4 is a regulator of lipid droplets. It is up-regulated in MPTP-induced mouse models of Parkinson's disease (PD), with PLIN4 levels resulting in disrupted mitophagy as well as impaired chaperone-mediated autophagy, a mechanism of protein quality control [31, 32]. PLIN4 has also been reported to be increased in disease-associated oligodendrocytes in a mouse model of amyotrophic lateral sclerosis [33]. While direct evidence of the role of PLIN4 in human neurodegenerative diseases is limited, lipid dysregulation is a hallmark of AD [34, 35]. Our findings indicate that astrocyte-directed APOE3Ch expression decreases PLIN4 likely in oligodendrocytes, suggesting that APOE3Ch may influence lipid handling in oligodendrocytes although further study is necessary. Moreover, recent observations support activation of ferroptosis, an iron-dependent form of programmed cell death driven by lipid peroxidation, in AD [36, 37]. Although evidence remains limited, emerging data has linked PLIN4 to ferroptosis with increased PLIN4 expression increasing ferroptosis in tumor cells [38], Leydig cells [39], and hippocampal neurons [40]. A recent study demonstrated that APOE3Ch microglia derived from human iPSC led to resistance to Aβ-induced lipid peroxidation and resulting ferroptosis [41]. The

reduction in PLIN4 that we observed with APOE3Ch astrocytic expression in particular may reflect decreased lipid peroxidation and potentially lessened activation of ferroptotic pathways in the context of amyloid pathology that is like that of the APOE3 group. Indeed, CE(20:4)-lipidated APOE3Ch induced significantly attenuated lipofuscinosis in neuronal cells compared to those of APOE3 [17]. APOE-mediated lipid uptake can influence lipid droplet storage and lipotoxicity, suggesting that altered lipid handling may be a central pathway in AD pathogenesis.

Mounting evidence from human-based, in vitro, and in vivo studies has linked APOE3Ch to reduction in tau pathology [3–7, 24, 26, 41, 42]. Our 5xFAD model lacks human tau pathology, which is a caveat of this amyloid model mice. We did not find any influence of astrocytic APOE overexpression on tau aggregation or phosphorylation at threonine 231, which is an early marker of AD tau pathology in human CSF [43]. Nonetheless, our findings suggest that APOE3Ch exerts modest protective functions over APOE3 in reducing soluble, neurotoxic Aβ species and their associated toxicity. This is in line with clinical observations in the original APOE3Ch homozygote carrier, who exhibited limited neurodegeneration despite high Aβ load [3]. The pronounced gliosis and neuritic dystrophy observed in the GFP group, despite oligomeric Aβ levels comparable to the APOE3Ch group, may reflect the toxic potential of diffusely aggregated Aβ. APOE has been shown to influence Aβ plaque morphology. *Apoe* knockout mice exhibit markedly reduced amyloid deposition and develop diffuse, amorphous plaques. [44–46]. Nonetheless, APOE deficiency likely exacerbates neuritic dystrophy around fibrillar plaques in APPPS1ΔE9 mice [46]. We observed that astrocytic expression of APOE3 and APOE3Ch reduced plaque burden and yielded more compact plaques, as assessed by plaque sphericity [14, 15] and mean X-34 intensity, with APOE3Ch showing the stronger effect. This could be due to increased levels of the APOE protein itself, or a combination of APOE levels and differential effects of mouse versus human APOE [47]. Moreover, plaque-associated BACE1, a marker of dystrophic neurites [48], was reduced with APOE3Ch relative to GFP controls. These results suggest that APOE3Ch promotes a form of plaque compaction that is less damaging to surrounding neurites, potentially by sequestering oligomeric Aβ species and dampening inflammatory and stress-related pathways, as indicated by our transcriptomic analyses. Since increased neuritic dystrophy has been shown to facilitate tau spreading in tau pre-formed fibrils injection models [49], the ability of APOE3Ch to reduce neuritic dystrophy may also help explain its protective effect against tau pathology, as observed in the *APOE3Ch* homozygous carrier with *PSEN1*-p.E280A mutation.

Multiple studies have demonstrated that *APOECh* mutation broadly suppresses gliosis, dampens inflammatory gene expression, and modulates disease-associated microglial (DAM) phenotypes in an amyloid and tau-context dependent manner [4, 5, 7, 42, 50]. For instance, APOE3Ch knockin led to overall decrease in both astrocytosis and microglial activation and, in parallel, increased CD68-positive microglial clustering compared with APOE3 knockin in APP/PS1 mice [4]. Moreover, knockin of APOECh into murine *Apoe* also enhances microglial recruitment to plaques and promoted a DAM phenotype in 5xFAD mice, concurrent with reduced plaque burden and neuritic injury [7]. While we also observe that astrocytic APOE3Ch expression suppresses both astrocytosis and IBA1-positive microglial activation in the cortex, it does so to a similar extent as astrocytic APOE3 expression. Consistent with our results, the AAVrh.10-mediated APOE2 and APOE2Ch expressions in the central nervous system results in a similar suppression of gliosis in APP/PS1 mice with *APOE4* [5]. Additionally, we observed reduced microglial recruitment to plaques or proximity to fibrillar deposits by astrocytic expression of APOE3 and APOE3Ch without differences between them. This observation is in contradiction with reports of increased microglial engagement to plaques [4, 7]. Studies have shown that APOE facilitates microglial response to plaques, with APOE deficiency resulting in decreased plaque compaction and reduced clustering of microglia around plaques [46]. Taking together with the available literature, our findings suggest that APOE3Ch expression in microglia rather than astrocytes is necessary to drive enhanced plaque-associated microglial phenotypes described in the knockin mouse models, reinforcing the importance of cell type-specific expression in shaping APOE-mediated modulation of the innate immune response.

Several limitations of this study should be considered. First, while we used a *GFAP* promoter to restrict transgene expression to astrocytes, AAV8 is known to have some degree of promoter-independent leakiness. In our study, although we did not observe GFP immunoreactivity in neurons in control mice, we noted a small number of APOE-HA positive cells with neuronal morphology in a subset of animals. This sparse expression likely reflects minor leakiness or variability in injection depth near the ventricular zone. It is also possible that neurons take up astrocyte-secreted APOE. Second, our analysis at 8 months of age captures the 5xFAD model at an advanced stage of pathology, which may limit detection of early or more dynamic APOE effects on plaque seeding or glial transitions. Finally, the 5xFAD model lacks human tau pathology and APOE4, which may not fully reflect the complexity of AD. Thus, our future studies should further investigate the therapeutic effects of APOE3Ch expression using amyloid and tau mouse models with APOE4 at different ages. Altogether, our findings expand upon recent reports characterizing APOE3Ch as a protective variant associated with familial AD cases carrying *PSEN1*. Here, we showed that astrocytic APOE3Ch expression modulates specific Aβ pools, reduces plaque-associated neuritic damage, and dampens microglial activation – even in the presence of endogenous murine *Apoe*. While differences between APOE3Ch and APOE3 were modest, the selective reduction in oligomeric Aβ with APOE3Ch suggests variant-specific biological activity. This is particularly important in the context of tau pathology, with several studies reporting that Aβ oligomers can induce tau hyperphosphorylation [51], mislocalization to dendrites and spines causing neurotoxicity [52], and increase binding and internalization of tau oligomers into synapses, facilitating tau spread [53]. This may be one of the mechanisms through which APOE3Ch may lead to decreased tau pathology, in addition to other mechanisms that have been proposed such as direct interaction of APOE3Ch with tau [24]. While astrocytic APOE3Ch can shift Aβ species, it does not fully recapitulate the microglial phenotype observed in the knockin mouse models, pointing to a potential role for microglia-derived APOECh or a cell-autonomous effect of APOECh on microglial response to Aβ. Since previous reports have highlighted structural differences between astrocytic and microglia derived APOE particles [54], our results further underscore the need for cell-type-specific interrogation of APOE biology. Although our AAV strategy offers valuable mechanistic insights into the astrocyte-specific actions of APOE3Ch, further optimizations would be necessary for therapeutic translation.

DATA AVAILABILITY

The bulk RNA-seq data that supports the findings of this study are deposited in the Gene Expression Omnibus repository under accession number GSE319850. All other data are available from the corresponding author upon reasonable request.

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AUTHOR CONTRIBUTIONS

A.C.R, G.B., and T.K. conceptualized and designed the study. A.C.R., G.B., and T.K. interpreted data and led the drafting of the manuscript. M.L. provided the AAV-GFAP-GFP/APOE3/APOE3Ch viruses. C.L. and A.A. performed intracranial injections and maintained animal colonies. A.C.R., A.A., C.L., S.V.D. harvested mice. A.C.R., A.A. and L.A.K. processed tissues for protein lysates. A.C.R. performed molecular biochemical assays and immunostaining and fluorescence imaging. P.R.M. assisted with fluorescence imaging and file conversion. A.R.Y. and B.R. assisted with confocal imaging. E.L. performed the aggregation assay. Y.R. performed bioinformatic analysis of bulk-RNA sequencing. A.C.R. performed data analysis including Imaris analysis. C.C.L., Y.A.M., G.B., and T.K. assisted with data analysis and interpretation. All authors contributed to writing and editing the final manuscript.

COMPETING INTERESTS

G.B. serves as a consultant for SciNeuro Pharmaceuticals. Y.A.M. is a current employee of SciNeuro Pharmaceuticals. C.-C.L. is a current employee of Simcere Innovation Center. All other authors declare no competing interest.

ETHICS

This study did not include human participants or human tissue and therefore does not need informed consent.

ADDITIONAL INFORMATION

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